

IS2205 – Mobile Application Design and Development

About the course

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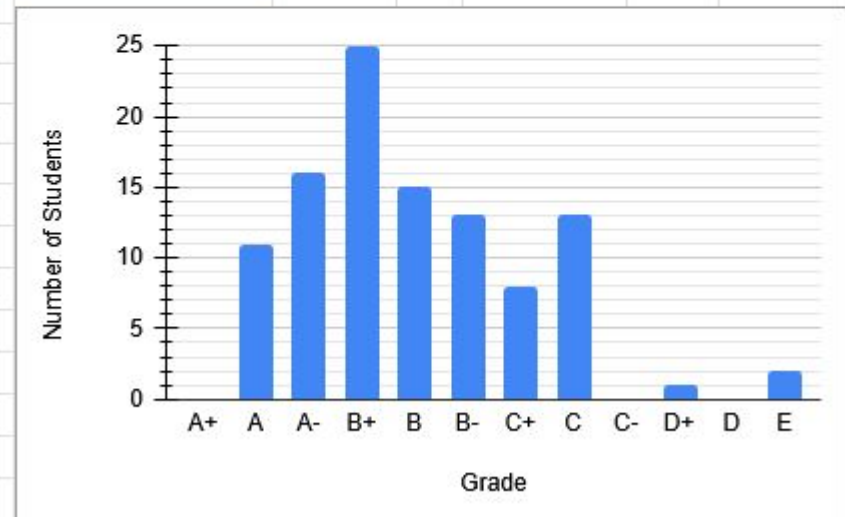
Course Structure

- ❑ Part 1 – Android App Development
- ❑ Part 2 – Other App Development Frameworks
- ❑ Assessment: Course work : 30% , Final exam : 70%
- ❑ Course work : QnA + Practicals + Mini Project
- ❑ Paper Format : 4 Mandatory Questions (Usually)

Last Semester

IS2205 - MADD

General Statistics		Fail	
Average Marks	57	Absent	6
Standard Deviation	12	NR	8
Max	83	Pass	101
Min	6	Pass Rate	97.12%
Median	60	Total Present	104
Mode	66		



Agenda

- ❑ What is mobile computing?
- ❑ Mobile communication
- ❑ Mobile hardware
- ❑ Mobile software
- ❑ Mobile application development
- ❑ Android platform architecture

What is Mobile Computing?

- ❑ A technology that facilitates processing & transmission of data via a computer or any other wireless enabled device without having to be stationed or connected to a fixed physical link
- ❑ Allow users to access information from portable devices
- ❑ Connected mode and disconnected mode
- ❑ Wired and wireless mode : Some mobile devices have physical network connectivity

Main Ingredients of Mobile Computing

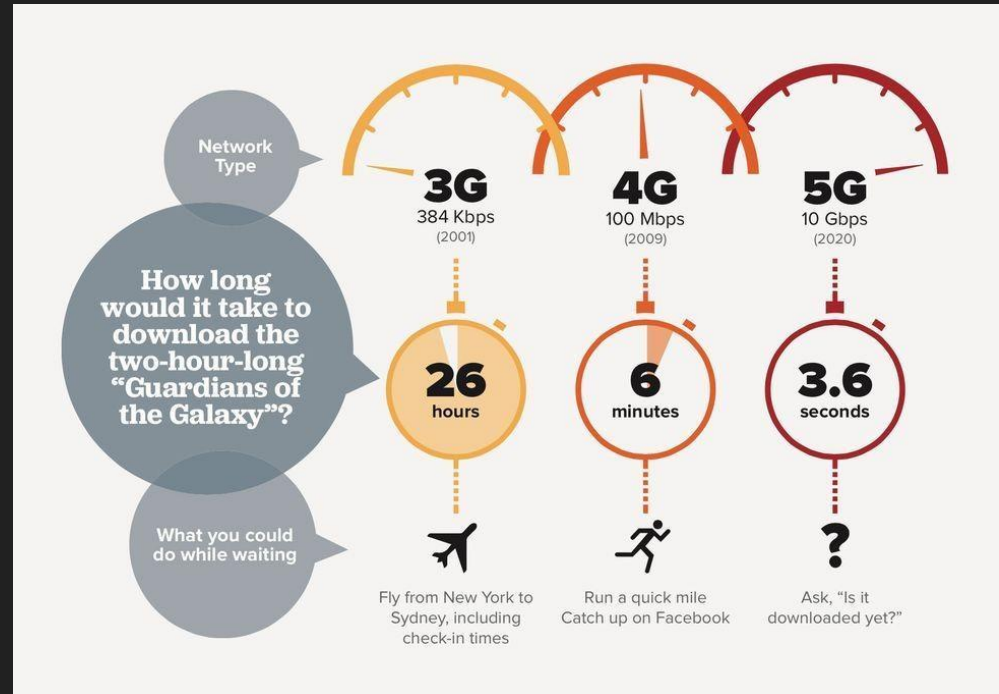
- ❑ Mobile Communication
- ❑ Mobile Hardware
- ❑ Mobile Software

Mobile Communication

- ❑ Mobile data & voice services
- ❑ Wi-Fi
- ❑ IR
- ❑ Bluetooth
- ❑ RFID
- ❑ GPS

Progress in Mobile Communication

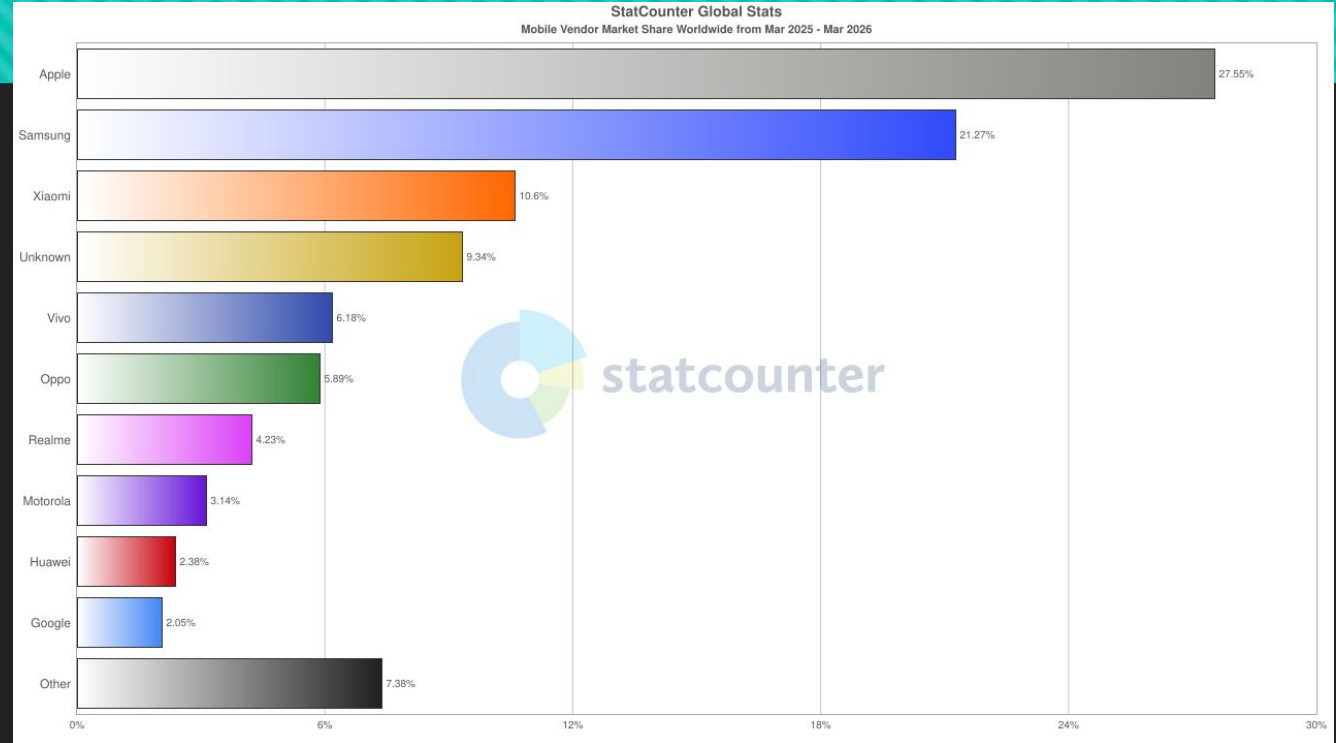
- Progress of mobile network speeds



Mobile Hardware

Mobile Device Market

Smartphone Market Share



Mobile Device Market...

Smartwatch Market Share

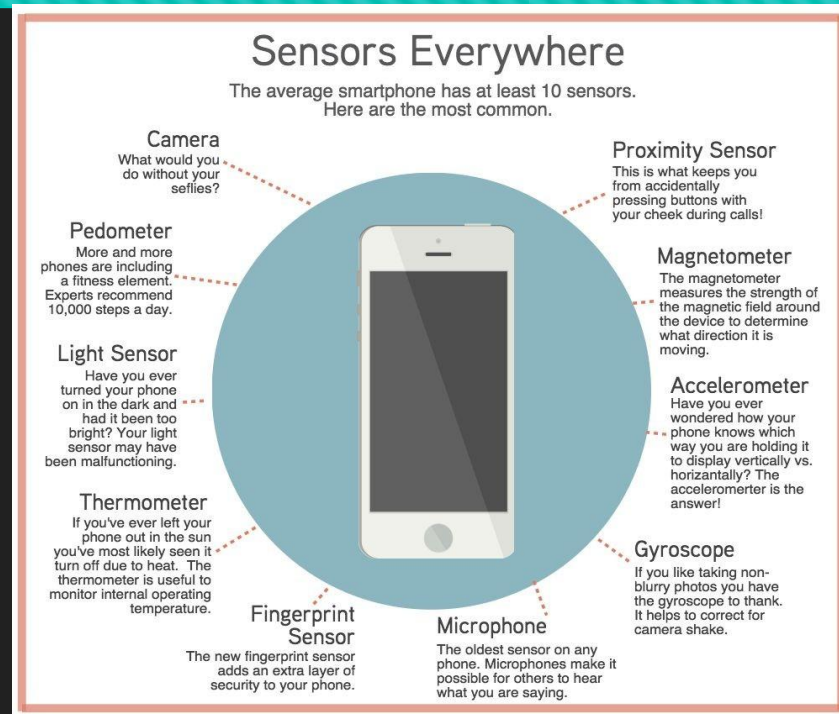


Mobile hardware...

- ❑ Mobile phone is no longer considered as just a communication device, rather it provides more powerful computing resources:
- ❑ High performance mobile processors + GPUs
- ❑ Large flash memory ~ 100GB
- ❑ Set of sensors

How many sensors

- Sensors play a crucial role in modern mobile devices



Activity

- ❑ How can we identify sensors in the mobile phone?
- ❑ Identify different sensors available in your own mobile phone and write a use-case for each of them

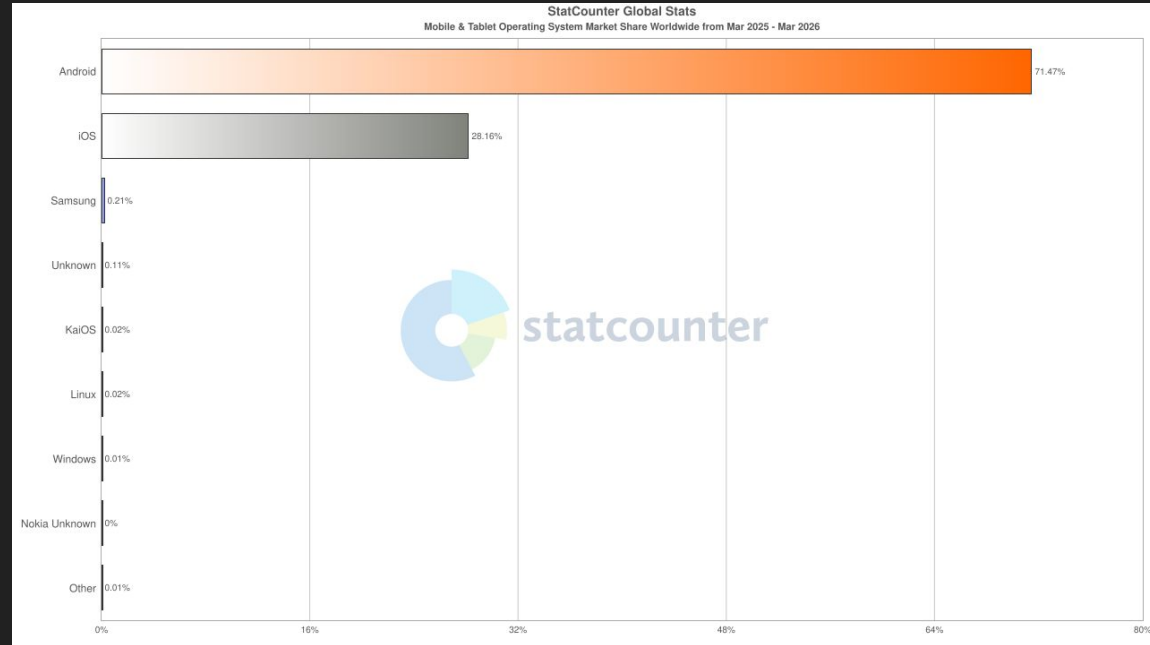
Mobile Software

Guess the global OS market share?

- ☐ Android ?%
- ☐ iOS ?%
- ☐ Others ?%

Mobile & Tablet Operating Systems - Global

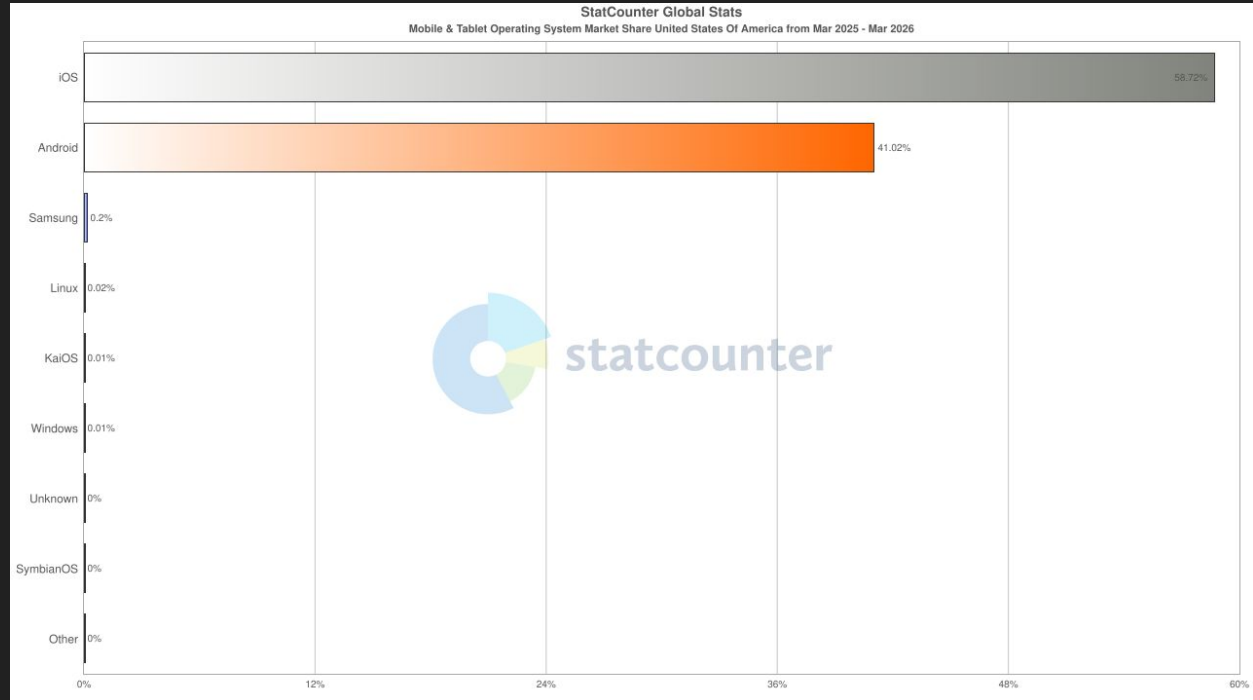
- Android ~ 71%
- iOS ~ 28%
- Others ~ 1%



Android & iOS - two market segments

- ❑ Many independent app developers choose to build their apps for Android first.
- ❑ Google Play Store has fewer restrictions compared to the Apple App Store.
- ❑ Applications developed for iOS have far fewer devices to support, making them perform well, compatible etc.
- ❑ User retention is typically higher for iOS.

Android & iOS - USA statistics



Question:

What specific reasons for USA mobile market being dominated by iOS?

Mobile Application Development

What is mobile application development?

- ❑ Mobile application development is the collection of operations involved for developing software running on mobile computing devices

Mobile Application Development...

- ❑ Both web application development & mobile application development has its roots in more traditional software development.
- ❑ However, mobile apps are often written specifically to take advantage of the unique features of a particular mobile device.
- ❑ A gaming app might be written to take advantage of the phone's accelerometer, a mobile health app might be written to take advantage of a temperature sensor in a smartwatch, a shopping cart app may use the device's barcode reader to find the information of a product.

Mobile Application Types

- ❑ Native Applications
- ❑ Web Applications
- ❑ Hybrid Applications

Native

NATIVE APPS



TECHNOLOGY USED

Java, Kotlin, Python, Swift, Objective C, etc.



PROS

- 1 Faster, better performance
- 2 Native UI
- 3 Can access device features



CONS

- 1 Higher cost to maintain
- 2 Takes up space in the device
- 3 Updates must be downloaded



WEB APPS



TECHNOLOGY USED

HTML5, CSS, JavaScript, Ruby, etc.



PROS

- 1 Web-based so performs on all devices
- 2 Easier to maintain
- 3 Users don't run out of storage

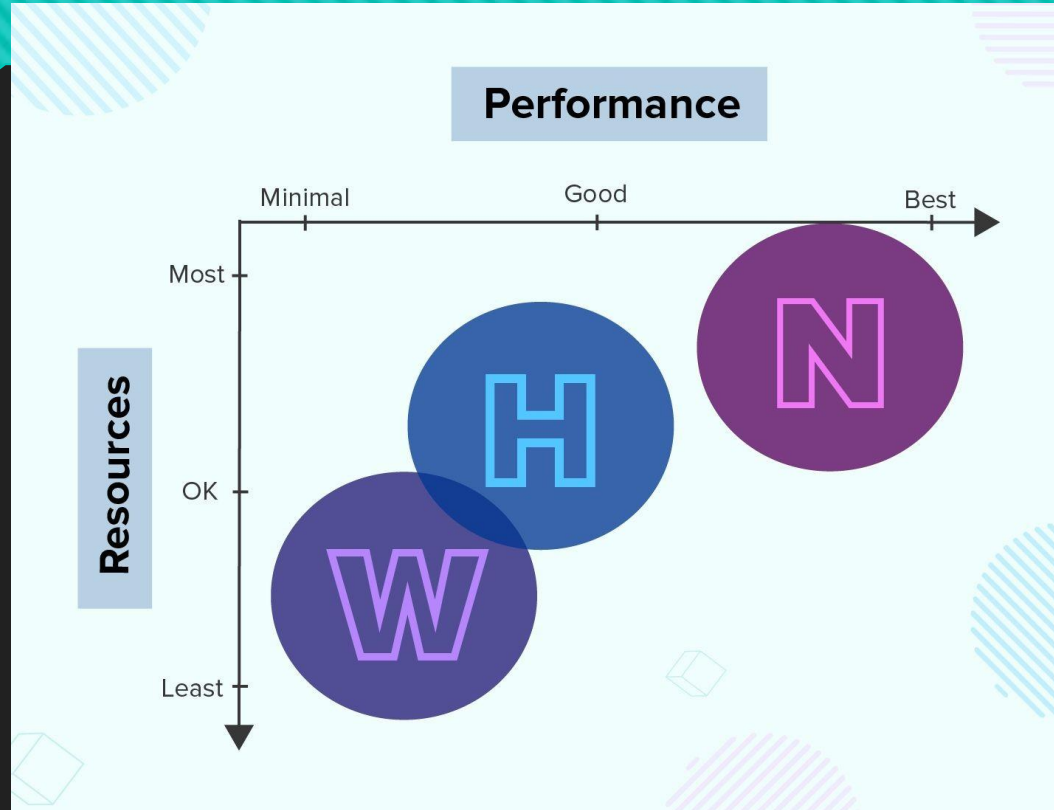


CONS

- 1 Dependent on a browser
- 2 Needs an internet connection
- 3 May not always integrate with device hardware



decrease



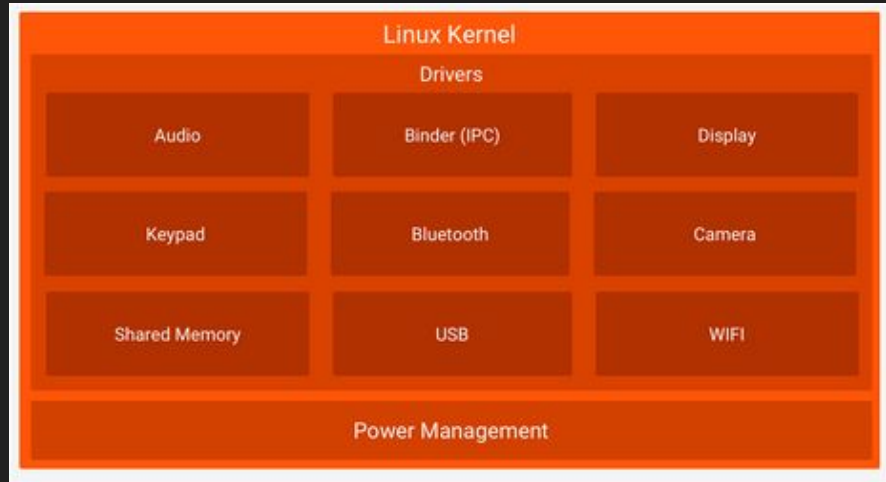
Increase

Question

Write a feature comparison between app types of native, web and hybrid in a table

Android Platform Architecture

- ❑ The foundation of the Android platform is the Linux kernel.
- ❑ Android Runtime (ART) relies on the Linux kernel for underlying functionalities such as threading and low-level memory management.
- ❑ Using a Linux kernel allows Android to take advantage of key security features and allows device manufacturers to develop hardware drivers for a well-known kernel



Android Platform Architecture...

- ❑ HAL provides standard interfaces that expose device hardware capabilities to the higher-level Java API framework.
- ❑ The HAL consists of multiple library modules, each of which implements an interface for a specific type of hardware component, i.e.: Camera or Bluetooth modules.
- ❑ When a framework API makes a call to access device hardware, the Android system loads the library module for that hardware component.



Android Platform Architecture...

- ❑ Android API level 21+, each app runs in its own instance of the Android Runtime (ART).
- ❑ ART runs by executing DEX files, a bytecode format designed specially for Android.
- ❑ Ahead-of-time (AOT) and just-in-time (JIT) compilation
- ❑ Garbage collection (GC)
- ❑ API level 28+, conversion of an app DEX files to more compact machine code.
- ❑ Better debugging support



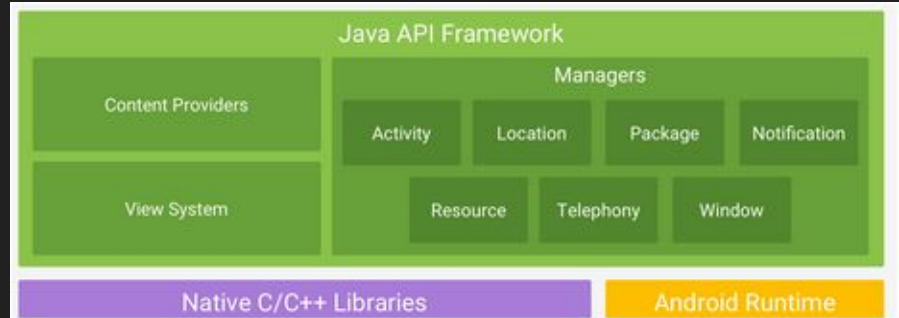
Android Platform Architecture...

- ❑ Many core Android system components and services, such as ART and HAL, are built from native code that require native libraries written in C and C++.
- ❑ The Android platform provides Java framework APIs to expose the functionality of some of these native libraries to apps.
- ❑ If your app requires C or C++ code, you can use the Android NDK to access some of these native platform libraries directly from your native code.



Android Platform Architecture...

- ❑ Bread and butter for app developers
- ❑ A rich and extensible View System you can use to build an app's UI
- ❑ Resource Manager, providing access to non-code resources such as localized strings, graphics, and layout files
- ❑ Notification Manager, Activity Manager...
- ❑ Content Providers that enable apps to access data from other apps or to share their own data



- ❑ Developers have full access to the same framework APIs that Android system apps use. (Most cases!)

Android Platform Architecture...

- ❑ Generally not different from developers' apps (Some exception such as settings app)
- ❑ The features available by these can be shared with your app.
- ❑ Do not reinvent the wheel but just ask for support!



Thank You!